COMPLETE LEARNING SESSION

# Arid Desert Landforms / Thar Desertification - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing


*Original deterministic study visual; labels are explanatory, not textual quotations.*

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

# CONTENTS / SESSION INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

# Arid Desert Landforms / Thar Desertification - Learner-v2 Complete Learning Session

> **Authoring-only generation:** 2026-09-01. No PDF was rendered and no tracker or index was mutated.

### SOURCE, PROGRESSION AND CURRENT-LINKAGE AUDIT

- **Generation date:** 2026-09-01.
- **Repository-first evidence:** the Basic owner is taught first and preserved in full; the Advanced owner is retained only in the optional final teaching block.
- **OCR evidence:** Repository Markdown was primary. The three required local Geography books were inspected as supplementary OCR/searchable evidence. No unsupported page precision, volatile figure or quotation was imported.
- **Qdrant:** not used; repository Markdown and available OCR context were sufficient.
- **PYQ integrity:** TRANSPARENT ZERO-DIRECT-PYQ AUDIT: the central Mains ledger routes the 2020 desertification demand to Environment and Ecology, and no direct Topic 07 objective route appears in the checked central ledgers. Cross-owned questions remain conceptual context, not claimed Geography PYQs.
- **Live-link boundary:** The SAC atlas page and UNCCD's June 2026 restoration communication were directly fetched. The atlas remains a dated mapping baseline, not a September 2026 condition report; no Thar percentage, target achievement or COP17 outcome is asserted.
- **Fact/inference discipline:** no current-affairs item, PYQ wording, figure or quotation is invented.

## BASIC LEARNING SESSION

### DEEP-REVIEW LEARNING CONTRACT

| Control | Binding rule for this package |
|---|---|
| Syllabus boundary | Complete physical process, spatial pattern and India anchor are taught before optional Advanced depth. |
| Causal method | Initial condition -> energy/force or driver -> mechanism -> landform/climate/ocean pattern -> consequence -> feedback/limit. |
| Spatial method | Latitude, altitude, continentality, relief, plate setting, basin/coast orientation and map scale are explicit. |
| Terminology | Close terms are defined on common axes; process, form, region, hazard, regulation and current status are never collapsed. |
| Evidence method | Claim -> named place/map/process/official record -> analysis -> qualification. |
| Practice contract | Every solved item has demand decoding, a detailed examiner-grade model, executable timed/compression plan, marks rationale and answer-specific improvement. |
| Approval | This immutable successor remains `approved: false` pending explicit approval. |

**Canonical Basic/Core owner:** `upsc-ai-kit\knowledge\Geography\basic\07_Arid-Desert-Landforms.md` **Canonical topic owner:** `upsc-ai-kit\knowledge\Geography\basic\07_Arid-Desert-Landforms.md` **Optional Advanced owner:** `upsc-ai-kit\knowledge\Geography\advanced\07_Thar-Desertification.md` **Official syllabus mapping:** `upsc-ai-kit\knowledge\Geography\OFFICIAL-UPSC-SYLLABUS-MAPPING.md`

## EVIDENCE, PYQ AND CURRENT-STATUS CONTROL

- **Process discipline:** no feature is listed without formation sequence, controlling variables and a limiting condition.
- **Map discipline:** distribution is reconstructed through latitude, plate/relief setting, wind/current direction, drainage/coast orientation and named anchors.
- **Quantitative discipline:** coordinates, thresholds, magnitudes, dates, counts and project dimensions retain units, source/date and uncertainty.
- **Causal discipline:** association is not treated as sufficiency; interacting controls, scale, lag, feedback and exceptions are stated.
- **PYQ discipline:** repository routing ledgers and held papers control wording and metadata; reconstructed wording and unavailable official keys remain labelled.
- **Status discipline:** every changeable claim retains an official source, observation date and status boundary.
- **Current-status note, rechecked 2026-09-05:** Rechecked 2026-09-05: SAC's 2021 Desertification and Land Degradation Atlas remains the latest located official nationwide mapping baseline. It reports 97.85 million hectares, or 29.77 percent of India's geographical area, under land degradation/desertification during 2018-19. MoEFCC's 2023 National Action Plan retains the 26-million-hectare restoration ambition for 2030, while UNCCD defines LDN through avoiding, reducing and reversing degradation. The atlas values are not relabelled as September 2026 conditions, a Thar-only rate or verified restoration achievement.

**Live/official context sources rechecked for this generation:**

- `https://vedas.sac.gov.in/vedas/dsm_atlas.html`
- `https://www.sac.gov.in/data/Publication/128/Desertification_and_Land_Degradation_Atlas_of_India_2021.pdf`
- `https://pib.gov.in/PressReleseDetailm.aspx?PRID=1727987`
- `https://moef.gov.in/uploads/2023/07/NAP%20final-2023.pdf`
- `https://www.unccd.int/land-and-life/land-degradation-neutrality/overview`


*Refreshed teaching navigation*

*Distinct embedded teaching-navigation image. The separate continuous Cārvāka-style flowchart package remains an independent at-a-glance artifact.*

# SESSION 1 - FOUNDATION - Desert, aridity, drought and degradation

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Desert, aridity, drought and degradation explains how Concept boundary shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, Desert, aridity, drought and degradation is the source-bounded relation among Concept boundary, classified by process, form, spatial setting, temporal change and human consequence.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Desert, aridity, drought and degradation must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

## MUST-WRITE KEYWORDS

- **Desert**
- **aridity**
- **drought**
- **degradation**
- **Concept**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - These terms are related but non-equivalent. - and conclude: Open the 2020 route with a definition firewall.

## VISUAL FIRST

```
DESERT, ARIDITY, DROUGHT AND DEGRADATION
01. Concept boundary
BOUNDARY -> These terms are related but non-equivalent.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

## CORE EXPLANATION

A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation.

## NAMED EVIDENCE AND MECHANISM

- **Concept boundary:** A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation.

## EXAMINER CAUTION

- These terms are related but non-equivalent.

## EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Desert, aridity, drought and degradation.
- **Mains:** Open the 2020 route with a definition firewall.

## MINI RECAP

- **Evidence chain:** Concept boundary
- **Qualified use:** Open the 2020 route with a definition firewall.

## CLOSING RECALL FLOW

**SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW**

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| Desert \| aridity \| drought \| degradation \| Concept | connect Concept boundary through process, place and time | Open the 2020 route with a definition firewall. | These terms are related but non-equivalent. |

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Desert, aridity, drought and degradation converts physical process into a qualified spatial argument

# SESSION 2 - FOUNDATION - Why deserts form

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Why deserts form explains how Aridity controls shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, Why deserts form is the source-bounded relation among Aridity controls, classified by process, form, spatial setting, temporal change and human consequence.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Why deserts form must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

## MUST-WRITE KEYWORDS

- **deserts**
- **form**
- **Aridity**
- **controls**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - High temperature alone does not define desert genesis. - and conclude: Classify deserts by atmospheric and locational control.

## VISUAL FIRST

```
WHY DESERTS FORM
01. Aridity controls
BOUNDARY -> High temperature alone does not define desert genesis.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

## CORE EXPLANATION

Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient.

## NAMED EVIDENCE AND MECHANISM

- **Aridity controls:** Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient.

## EXAMINER CAUTION

- High temperature alone does not define desert genesis.

## EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Why deserts form.
- **Mains:** Classify deserts by atmospheric and locational control.

## MINI RECAP

- **Evidence chain:** Aridity controls
- **Qualified use:** Classify deserts by atmospheric and locational control.

## CLOSING RECALL FLOW

**SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW**

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
| --- | --- | --- | --- |
| deserts \| form \| Aridity \| controls | connect Aridity controls through process, place and time | Classify deserts by atmospheric and locational control. | High temperature alone does not define desert genesis. |

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Why deserts form converts physical process into a qualified spatial argument

# SESSION 3 - FOUNDATION - Weathering and sediment supply

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Weathering and sediment supply explains how Weathering in arid lands shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, Weathering and sediment supply is the source-bounded relation among Weathering in arid lands, classified by process, form, spatial setting, temporal change and human consequence.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Weathering and sediment supply must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

## MUST-WRITE KEYWORDS

- **Weathering**
- **sediment**
- **supply**
- **arid**
- **lands**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - Arid weathering includes thermal, salt and episodic moisture effects. - and conclude: Connect material production to later transport.

## VISUAL FIRST

```
WEATHERING AND SEDIMENT SUPPLY
01. Weathering in arid lands
BOUNDARY -> Arid weathering includes thermal, salt and episodic moisture effects.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

## CORE EXPLANATION

Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff.

## NAMED EVIDENCE AND MECHANISM

- **Weathering in arid lands:** Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff.

## EXAMINER CAUTION

- Arid weathering includes thermal, salt and episodic moisture effects.

## EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Weathering and sediment supply.
- **Mains:** Connect material production to later transport.

## MINI RECAP

- **Evidence chain:** Weathering in arid lands
- **Qualified use:** Connect material production to later transport.

## CLOSING RECALL FLOW

**SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW**

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| Weathering \| sediment \| supply \| arid \| lands | connect Weathering in arid lands through process, place and time | Connect material production to later transport. | Arid weathering includes thermal, salt and episodic moisture effects. |

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Weathering and sediment supply converts physical process into a qualified spatial argument**

# SESSION 4 - FOUNDATION - Creep, saltation and suspension

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Creep, saltation and suspension explains how Aeolian transport shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, Creep, saltation and suspension is the source-bounded relation among Aeolian transport, classified by process, form, spatial setting, temporal change and human consequence.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Creep, saltation and suspension must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

## MUST-WRITE KEYWORDS

- **Creep**
- **saltation**
- **suspension**
- **Aeolian**
- **transport**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - Transport mode depends on particle size and wind threshold. - and conclude: Use process vocabulary for elimination.

## VISUAL FIRST

```
CREEP, SALTATION AND SUSPENSION
01. Aeolian transport
BOUNDARY -> Transport mode depends on particle size and wind threshold.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

## CORE EXPLANATION

Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces.

## NAMED EVIDENCE AND MECHANISM

- **Aeolian transport:** Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces.

## EXAMINER CAUTION

- Transport mode depends on particle size and wind threshold.

## EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Creep, saltation and suspension.
- **Mains:** Use process vocabulary for elimination.

## MINI RECAP

- **Evidence chain:** Aeolian transport
- **Qualified use:** Use process vocabulary for elimination.

## CLOSING RECALL FLOW

**SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW**

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| Creep \| saltation \| suspension \| Aeolian \| transport | connect Aeolian transport through process, place and time | Use process vocabulary for elimination. | Transport mode depends on particle size and wind threshold. |

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Creep, saltation and suspension converts physical process into a qualified spatial argument**

# SESSION 5 - CORE - Wind erosion and residual landforms

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Wind erosion and residual landforms explains how Erosional landforms shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, Wind erosion and residual landforms is the source-bounded relation among Erosional landforms, classified by process, form, spatial setting, temporal change and human consequence.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Wind erosion and residual landforms must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

## MUST-WRITE KEYWORDS

- **Wind**
- **erosion**
- **residual**
- **landforms**
- **Erosional**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - Many forms also require structural or weathering controls. - and conclude: Avoid wind-only explanations.

## VISUAL FIRST

```
WIND EROSION AND RESIDUAL LANDFORMS
01. Erosional landforms
BOUNDARY -> Many forms also require structural or weathering controls.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

## CORE EXPLANATION

Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering.

## NAMED EVIDENCE AND MECHANISM

- **Erosional landforms:** Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering.

## EXAMINER CAUTION

- Many forms also require structural or weathering controls.

## EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Wind erosion and residual landforms.
- **Mains:** Avoid wind-only explanations.

## MINI RECAP

- **Evidence chain:** Erosional landforms
- **Qualified use:** Avoid wind-only explanations.

## CLOSING RECALL FLOW

## SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| Wind \| erosion \| residual \| landforms \| Erosional | connect Erosional landforms through process, place and time | Avoid wind-only explanations. | Many forms also require structural or weathering controls. |

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Wind erosion and residual landforms converts physical process into a qualified spatial argument**

# SESSION 6 - CORE - Dune types and wind regimes

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Dune types and wind regimes explains how Dune anatomy shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, Dune types and wind regimes is the source-bounded relation among Dune anatomy, classified by process, form, spatial setting, temporal change and human consequence.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Dune types and wind regimes must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

### MUST-WRITE KEYWORDS

- **Dune**
- **types**
- **wind**
- **regimes**
- **anatomy**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - Barchan horns point downwind. - and conclude: Read dune shape as process evidence.

### VISUAL FIRST

```
DUNE TYPES AND WIND REGIMES
01. Dune anatomy
BOUNDARY -> Barchan horns point downwind.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

### CORE EXPLANATION

A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes.

### NAMED EVIDENCE AND MECHANISM

- **Dune anatomy:** A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes.

### EXAMINER CAUTION

- Barchan horns point downwind.

### EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Dune types and wind regimes.
- **Mains:** Read dune shape as process evidence.

### MINI RECAP

- **Evidence chain:** Dune anatomy
- **Qualified use:** Read dune shape as process evidence.

### CLOSING RECALL FLOW

## SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| Dune \| types \| wind \| regimes \| anatomy | connect Dune anatomy through process, place and time | Read dune shape as process evidence. | Barchan horns point downwind. |

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Dune types and wind regimes converts physical process into a qualified spatial argument**

# SESSION 7 - CORE - Wadis, fans, pediments and playas

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Wadis, fans, pediments and playas explains how Water-built desert forms shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, Wadis, fans, pediments and playas is the source-bounded relation among Water-built desert forms, classified by process, form, spatial setting, temporal change and human consequence.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Wadis, fans, pediments and playas must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

## MUST-WRITE KEYWORDS

- **Wadis**
- **fans**
- **pediments**
- **playas**
- **Water-built**
- **desert**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - Deserts are not geomorphically waterless. - and conclude: Integrate episodic runoff and evaporation.

## VISUAL FIRST

```
WADIS, FANS, PEDIMENTS AND PLAYAS
01. Water-built desert forms
BOUNDARY -> Deserts are not geomorphically waterless.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

## CORE EXPLANATION

Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology.

## NAMED EVIDENCE AND MECHANISM

- **Water-built desert forms:** Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology.

## EXAMINER CAUTION

- Deserts are not geomorphically waterless.

## EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Wadis, fans, pediments and playas.
- **Mains:** Integrate episodic runoff and evaporation.

## MINI RECAP

- **Evidence chain:** Water-built desert forms

- **Qualified use:** Integrate episodic runoff and evaporation.

## CLOSING RECALL FLOW

**SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW**

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| Wadis \| fans \| pediments \| playas \| Water-built \| desert | connect Water-built desert forms through process, place and time | Integrate episodic runoff and evaporation. | Deserts are not geomorphically waterless. |

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Wadis, fans, pediments and playas converts physical process into a qualified spatial argument**

# SESSION 8 - CORE - Thar regional mosaic

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Thar regional mosaic explains how Thar spatial mosaic and Thar climate and ecology shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, Thar regional mosaic is the source-bounded relation among Thar spatial mosaic and Thar climate and ecology, classified by process, form, spatial setting, temporal change and human consequence.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Thar regional mosaic must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

## MUST-WRITE KEYWORDS

- **Thar**
- **regional**
- **mosaic**
- **spatial**
- **climate**
- **ecology**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - A schematic region is not a legal boundary. - and conclude: Draw dunes, Aravalli, Luni, playas and canal tract.

## VISUAL FIRST

```
THAR REGIONAL MOSAIC
01. Thar spatial mosaic
    |
    v
02. Thar climate and ecology
BOUNDARY -> A schematic region is not a legal boundary.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

## CORE EXPLANATION

The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods.

## NAMED EVIDENCE AND MECHANISM

- **Thar spatial mosaic:** The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg.
- **Thar climate and ecology:** Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods.

## EXAMINER CAUTION

- A schematic region is not a legal boundary.

## EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Thar regional mosaic.
- **Mains:** Draw dunes, Aravalli, Luni, playas and canal tract.

## MINI RECAP

- **Evidence chain:** Thar spatial mosaic -> Thar climate and ecology
- **Qualified use:** Draw dunes, Aravalli, Luni, playas and canal tract.

## CLOSING RECALL FLOW

**SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW**

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| Thar \| regional \| mosaic \| spatial \| climate \| ecology | connect Thar spatial mosaic and Thar climate and ecology through process, place and time | Draw dunes, Aravalli, Luni, playas and canal tract. | A schematic region is not a legal boundary. |

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Thar regional mosaic converts physical process into a qualified spatial argument**

# SESSION 9 - CORE - Thar origin and susceptibility

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Thar origin and susceptibility explains how Origin debate and Natural susceptibility shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, Thar origin and susceptibility is the source-bounded relation among Origin debate and Natural susceptibility, classified by process, form, spatial setting, temporal change and human consequence.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Thar origin and susceptibility must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

## MUST-WRITE KEYWORDS

- **Thar**

- **origin**
- **susceptibility**
- **debate**
- **Natural**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - Origin and present degradation are different questions. - and conclude: Give a polygenetic, multi-timescale account.

## VISUAL FIRST

```
THAR ORIGIN AND SUSCEPTIBILITY
01. Origin debate
    |
    v
02. Natural susceptibility
BOUNDARY -> Origin and present degradation are different questions.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

## CORE EXPLANATION

The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation.

## NAMED EVIDENCE AND MECHANISM

- **Origin debate:** The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe.
- **Natural susceptibility:** Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation.

## EXAMINER CAUTION

- Origin and present degradation are different questions.

## EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Thar origin and susceptibility.
- **Mains:** Give a polygenetic, multi-timescale account.

## MINI RECAP

- **Evidence chain:** Origin debate -> Natural susceptibility
- **Qualified use:** Give a polygenetic, multi-timescale account.

## CLOSING RECALL FLOW

**SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW**

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| Thar \| origin \| susceptibility \| debate \| Natural | connect Origin debate and Natural susceptibility through process, place and time | Give a polygenetic, multi-timescale account. | Origin and present degradation are different questions. |

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Thar origin and susceptibility converts physical process into a qualified spatial argument**

# SESSION 10 - CORE - Human acceleration and thresholds

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Human acceleration and thresholds explains how Human accelerators shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, Human acceleration and thresholds is the source-bounded relation among Human accelerators, classified by process, form, spatial setting, temporal change and human consequence.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Human acceleration and thresholds must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

## MUST-WRITE KEYWORDS

- **Human**
- **acceleration**
- **thresholds**
- **accelerators**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - Presence of people is not itself degradation. - and conclude: Link pressure to loss of recovery capacity.

## VISUAL FIRST

```
HUMAN ACCELERATION AND THRESHOLDS
01. Human accelerators
BOUNDARY -> Presence of people is not itself degradation.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

## CORE EXPLANATION

Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds.

## NAMED EVIDENCE AND MECHANISM

- **Human accelerators:** Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds.

## EXAMINER CAUTION

- Presence of people is not itself degradation.

## EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Human acceleration and thresholds.
- **Mains:** Link pressure to loss of recovery capacity.

## MINI RECAP

- **Evidence chain:** Human accelerators
- **Qualified use:** Link pressure to loss of recovery capacity.

## CLOSING RECALL FLOW

**SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW**

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| Human \| acceleration \| thresholds \| accelerators | connect Human accelerators through process, place and time | Link pressure to loss of recovery capacity. | Presence of people is not itself degradation. |

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Human acceleration and thresholds converts physical process into a qualified spatial argument**

# SESSION 11 - CORE - Aravalli ecological functions

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Aravalli ecological functions explains how Aravalli function shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, Aravalli ecological functions is the source-bounded relation among Aravalli function, classified by process, form, spatial setting, temporal change and human consequence.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Aravalli ecological functions must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

### MUST-WRITE KEYWORDS

- **Aravalli**
- **ecological**
- **functions**
- **function**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - Do not use the simplistic wall metaphor. - and conclude: Explain relief, drainage and connectivity.

### VISUAL FIRST

```
ARAVALLI ECOLOGICAL FUNCTIONS
01. Aravalli function
BOUNDARY -> Do not use the simplistic wall metaphor.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

### CORE EXPLANATION

The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement.

### NAMED EVIDENCE AND MECHANISM

- **Aravalli function:** The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement.

### EXAMINER CAUTION

- Do not use the simplistic wall metaphor.

### EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Aravalli ecological functions.
- **Mains:** Explain relief, drainage and connectivity.

### MINI RECAP

- **Evidence chain:** Aravalli function
- **Qualified use:** Explain relief, drainage and connectivity.

### CLOSING RECALL FLOW

**SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW**

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| Aravalli \| ecological \| functions \| function | connect Aravalli function through process, place and time | Explain relief, drainage and connectivity. | Do not use the simplistic wall metaphor. |

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Aravalli ecological functions converts physical process into a qualified spatial argument**

# SESSION 12 - CORE - Indira Gandhi Canal trade-offs

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Indira Gandhi Canal trade-offs explains how Indira Gandhi Canal balance shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, Indira Gandhi Canal trade-offs is the source-bounded relation among Indira Gandhi Canal balance, classified by process, form, spatial setting, temporal change and human consequence.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Indira Gandhi Canal trade-offs must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

## MUST-WRITE KEYWORDS

- **Indira**
- **Gandhi**
- **Canal**
- **trade-offs**
- **balance**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - Benefits and salinity-waterlogging risks can coexist. - and conclude: Use a balanced irrigation transformation case.

## VISUAL FIRST

```
INDIRA GANDHI CANAL TRADE-OFFS
01. Indira Gandhi Canal balance
BOUNDARY -> Benefits and salinity-waterlogging risks can coexist.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

## CORE EXPLANATION

Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak.

## NAMED EVIDENCE AND MECHANISM

- **Indira Gandhi Canal balance:** Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak.

## EXAMINER CAUTION

- Benefits and salinity-waterlogging risks can coexist.

## EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Indira Gandhi Canal trade-offs.
- **Mains:** Use a balanced irrigation transformation case.

## MINI RECAP

- **Evidence chain:** Indira Gandhi Canal balance
- **Qualified use:** Use a balanced irrigation transformation case.

## CLOSING RECALL FLOW

**SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW**

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| Indira \| Gandhi \| Canal \| trade-offs \| balance | connect Indira Gandhi Canal balance through process, place and time | Use a balanced irrigation transformation case. | Benefits and salinity-waterlogging risks can coexist. |

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Indira Gandhi Canal trade-offs converts physical process into a qualified spatial argument**

# SESSION 13 - SYNTHESIS - PYQ boundary and LDN

## DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** PYQ boundary and LDN explains how Desertification beyond desert and LDN logic shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, PYQ boundary and LDN is the source-bounded relation among Desertification beyond desert and LDN logic, classified by process, form, spatial setting, temporal change and human consequence.

## ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> PYQ boundary and LDN must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

## MUST-WRITE KEYWORDS

- **Desertification**
- **beyond**
- **desert**
- **logic**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - Broad wording does not erase formal dryland usage. - and conclude: Reconcile exam demand with UNCCD precision.

## VISUAL FIRST

```
PYQ BOUNDARY AND LDN
01. Desertification beyond desert
     |
     v
02. LDN logic
BOUNDARY -> Broad wording does not erase formal dryland usage.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

## CORE EXPLANATION

The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital.

## NAMED EVIDENCE AND MECHANISM

- **Desertification beyond desert:** The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation.
- **LDN logic:** Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital.

## EXAMINER CAUTION

- Broad wording does not erase formal dryland usage.

## EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to PYQ boundary and LDN.
- **Mains:** Reconcile exam demand with UNCCD precision.

## MINI RECAP

- **Evidence chain:** Desertification beyond desert -> LDN logic
- **Qualified use:** Reconcile exam demand with UNCCD precision.

## CLOSING RECALL FLOW

## SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| Desertification \| beyond \| desert \| logic | connect Desertification beyond desert and LDN logic through process, place and time | Reconcile exam demand with UNCCD precision. | Broad wording does not erase formal dryland usage. |

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PYQ boundary and LDN converts physical process into a qualified spatial argument**

# SESSION 14 - SYNTHESIS - Restoration and monitoring

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** Restoration and monitoring explains how Process-matched restoration and Monitoring discipline shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, Restoration and monitoring is the source-bounded relation among Process-matched restoration and Monitoring discipline, classified by process, form, spatial setting, temporal change and human consequence.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Restoration and monitoring must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

### MUST-WRITE KEYWORDS

- **Restoration**
- **monitoring**
- **Process-matched**
- **discipline**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - A map is a baseline, not causal proof. - and conclude: Match intervention and indicator to process.

### VISUAL FIRST

```
RESTORATION AND MONITORING
01. Process-matched restoration
    |
    v
02. Monitoring discipline
BOUNDARY -> A map is a baseline, not causal proof.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

### CORE EXPLANATION

Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact.

### NAMED EVIDENCE AND MECHANISM

- **Process-matched restoration:** Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process.
- **Monitoring discipline:** Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact.

### EXAMINER CAUTION

- A map is a baseline, not causal proof.

### EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Restoration and monitoring.
- **Mains:** Match intervention and indicator to process.

### MINI RECAP

- **Evidence chain:** Process-matched restoration -> Monitoring discipline
- **Qualified use:** Match intervention and indicator to process.

### CLOSING RECALL FLOW

**SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW**

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| Restoration \| monitoring \| Process-matched \| discipline | connect Process-matched restoration and Monitoring discipline through process, place and time | Match intervention and indicator to process. | A map is a baseline, not causal proof. |

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Restoration and monitoring converts physical process into a qualified spatial argument**

## SESSION 15 - SYNTHESIS - PYQ ownership and current boundary

### DEFINITION / WHAT THIS IS CALLED

**Plain-language definition:** PYQ ownership and current boundary explains how Verified PYQ boundary and Current evidence boundary shape the examinable geography of Arid Desert Landforms / Thar Desertification.

**Technical definition:** In geographical analysis, PYQ ownership and current boundary is the source-bounded relation among Verified PYQ boundary and Current evidence boundary, classified by process, form, spatial setting, temporal change and human consequence.

### ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> PYQ ownership and current boundary must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

### MUST-WRITE KEYWORDS

- **ownership**
- **Verified**
- **classification**
- **causation**
- **comparison**
- **qualification**

**How to use them:** Define the process, locate its spatial expression, cite named evidence, apply this limit - Do not claim a cross-owned demand as a direct Geography PYQ. - and conclude: Close with definition, diagnosis, response and caveat.

## VISUAL FIRST

```
PYQ OWNERSHIP AND CURRENT BOUNDARY
01. Verified PYQ boundary
    |
    v
02. Current evidence boundary
BOUNDARY -> Do not claim a cross-owned demand as a direct Geography PYQ.
```

*This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.*

## CORE EXPLANATION

The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. SAC's 2021 atlas reports 97.85 million hectares, or 29.77 percent of India's geographical area, under land degradation/desertification in 2018-19. This is a dated national mapping baseline, not a September 2026 condition report or a Thar-only percentage.

## NAMED EVIDENCE AND MECHANISM

- **Verified PYQ boundary:** The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only.
- **Current evidence boundary:** SAC's 2021 atlas reports 97.85 million hectares, or 29.77 percent of India's geographical area, under land degradation/desertification in 2018-19. This is a dated national mapping baseline, not a September 2026 condition report or a Thar-only percentage.

## EXAMINER CAUTION

- Do not claim a cross-owned demand as a direct Geography PYQ.

## EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to PYQ ownership and current boundary.
- **Mains:** Close with definition, diagnosis, response and caveat.

## MINI RECAP

- **Evidence chain:** Verified PYQ boundary -> Current evidence boundary
- **Qualified use:** Close with definition, diagnosis, response and caveat.

## CLOSING RECALL FLOW

**SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW**

| 1. KEY TERMS / DEFINITIONS | 2. MECHANISM / ARGUMENT | 3. CONSEQUENCE / CONTRAST | 4. UPSC TRAP / ANSWER-USE |
|---|---|---|---|
| ownership \| Verified \| classification \| causation \| comparison \| qualification | connect Verified PYQ boundary and Current evidence boundary through process, place and time | Close with definition, diagnosis, response and caveat. | Do not claim a cross-owned demand as a direct Geography PYQ. |

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PYQ ownership and current boundary converts physical process into a qualified spatial argument**

## COMPLETE BASIC OWNER EVIDENCE BANK

> **Subject:** Geography · **Tier:** Must-Do (foundation + standard) · **GS Paper:** GS-I **Grounded in:** G.C. Leong, *Certificate Physical & Human Geography* + Majid Husain, *Indian & World Geography*. FACT: = from source book · ANALYSIS: = inference / standard knowledge · CURRENT: = current affairs. *Companion:* `advanced/07_Thar-Desertification.md`.

### 1. Snapshot & core idea

**Foundation - Wind Erosion & Depositional Desert Landforms (GC Leong / Majid).**

FACT: In deserts, wind is the chief agent. It erodes by deflation (lifting/rolling loose material) and abrasion (sand-blasting rock), then deposits sand as dunes and fine dust as loess. Sparse vegetation + dry, loose regolith let wind pick up grains. Coarse sand does the abrading near the ground; fine dust travels far and settles as loess beyond desert margins.

- FACT: Deflation lowers the land into deflation hollows by blowing away loose sand/dust; a lag of pebbles left behind forms a desert pavement.
- FACT: Abrasion and deflation help sculpt **zeugens** (tabular, resistant cap rock over a narrower softer

pedestal), pedestal/mushroom rocks and **yardangs** (streamlined ridges parallel to prevailing wind).

- FACT: Ventifacts are wind-faceted pebbles; a three-faced one is a dreikanter. Inselbergs are isolated

residual hills. Mesas and buttes are structurally controlled residuals shaped by combined weathering, runoff and slope retreat-not wind-only forms.

- FACT: Depositional dunes: crescent-shaped barchans (horns point downwind) and long ridge-like seifs (longitudinal dunes) aligned with the wind.
- FACT: Loess = extensive deposits of wind-blown fine silt/dust carried far beyond the desert (very fertile).

**Ca Anchor - Greening of the Thar Desert.**

FACT: The Thar shows that a 'desert' is dynamic: aeolian dune landscapes can be partly stabilised/greened by irrigation and land-use change, while other margins still face wind erosion.

- FACT: Drivers of local greening include Indira Gandhi Canal irrigation, groundwater use, farming,

settlement growth and rainfall variability; avoid transferring study-area percentage changes to the whole Thar.

- FACT: Greening stabilises dunes locally, but the arid fringe (E/SE Thar) remains vulnerable to degradation.
- FACT: Shows the difference between the physical 'sand desert' and productivity-based 'desertification'.

## 2. Key classification / data

| Process / landform | Formation | Diagnostic feature |
|---|---|---|
| FACT: Deflation | Wind blows away loose sand and dust | Deflation hollow; desert pavement lag |
| FACT: Abrasion | Sand-blasting by wind-driven grains | Zeugen, yardangs, ventifacts, dreikanter |
| FACT: Yardang | Streamlined wind-eroded ridge | Long axis parallel to prevailing wind |
| FACT: Zeugens | Differential erosion beneath a resistant cap | Tabular cap on a narrower pedestal; not every mushroom rock is a zeugen |
| FACT: Ventifact / dreikanter | Wind-faceted pebble | Dreikanter has three facets |
| FACT: Inselberg / mesa | Resistant residual hill / flat-topped upland | Left after surrounding softer material is removed |
| FACT: Barchan | Crescentic sand dune | Horns point downwind |
| FACT: Seif | Longitudinal dune | Long ridge aligned with dominant wind |
| FACT: Loess | Wind-blown fine silt deposited beyond desert margins | Often porous and fertile; colour and carbonate content vary |

## 3. Study links

> **Study link:** FACT: Geography -> Physical -> Arid/Aeolian Landforms (GC Leong: Wind & Deserts; Majid: Aeolian Geomorphology) **Study link:** FACT: Geography -> Arid Landforms -> Desert Dynamics (applied CA)

## 4. Must-Know Facts (Prelims)

**Foundation - Wind Erosion & Depositional Desert Landforms (GC Leong / Majid).**

- FACT: Wind erosion = deflation + abrasion.
- FACT: Deflation hollow + desert pavement (lag of pebbles).
- FACT: Yardang: long axis PARALLEL to wind; zeugen = rock mushroom.
- FACT: Dreikanter = 3-faced wind-faceted pebble (ventifact).

- FACT: Barchan = crescentic dune; seif = longitudinal dune.
- FACT: Loess = wind-blown fertile silt beyond desert margins.

**Ca Anchor - Greening of the Thar Desert.**

- FACT: Thar greening driven mainly by Indira Gandhi Canal irrigation.
- FACT: It is unusually densely settled for a hot desert, but avoid an uncited world superlative.
- FACT: Desert extent is dynamic - greening vs fringe degradation.

## 5. UPSC Traps

> MEMORY: Trap: Do not confuse similar-looking landforms, processes, scales or Indian locations; UPSC often tests the exact mechanism and setting.

**Foundation - Wind Erosion & Depositional Desert Landforms (GC Leong / Majid).**

- WRONG: Barchan horns point into (upwind of) the wind. -> Barchan horns point DOWNWIND (leeward); the gentle slope faces the wind.
- WRONG: A yardang's long axis is perpendicular to the wind. -> A yardang is streamlined PARALLEL to the wind direction.
- WRONG: Loess is a water-laid deposit. -> Loess is WIND-blown fine dust (aeolian), not fluvial.

**Ca Anchor - Greening of the Thar Desert.**

- WRONG: Deserts are static and always expanding. -> Extent is dynamic - irrigation can green/stabilise dunes while other areas degrade.

## 6. CURRENT: Current link

**Foundation - Wind Erosion & Depositional Desert Landforms (GC Leong / Majid).** CURRENT: Desert (aeolian) geomorphology underpins India's arid Thar - where wind erosion, dunes and desertification meet irrigation-driven greening.

**Ca Anchor - Greening of the Thar Desert.** CURRENT: Remote-sensing studies show parts of the Thar 'greening' over two decades - driven by canal irrigation (Indira Gandhi Canal), farm expansion and rising rainfall - even as fringes still degrade.

## 7. Mains angles

**Foundation - Wind Erosion & Depositional Desert Landforms (GC Leong / Majid).**

- ANALYSIS: Explain deflation vs abrasion and classify desert landforms, applying them to India's Thar (dunes, wind erosion) and its dynamic greening/desertification.

**Ca Anchor - Greening of the Thar Desert.**

- ANALYSIS: Use aeolian geomorphology to discuss irrigation-driven Thar greening vs desertification trade-offs.

## 8. Why deserts are where they are - and the work of water in them

> **Why this section exists:** the file taught aeolian landforms but never explained **why arid zones occupy the positions they do**, and it omitted the fluvial landform suite that in fact dominates most desert surfaces. Both are standard demands and both were absent.

### 8.1 The five causes of aridity, each with its signature

| Cause | Mechanism | Signature location |
|---|---|---|
| ANALYSIS: Subtropical high pressure | Descending limb of the Hadley circulation; subsidence warms and dries the air and suppresses convection | The great trade-wind deserts along roughly 20-30 degrees in both hemispheres - Sahara, Arabian, Kalahari, Australian |
| ANALYSIS: Continentality | Distance from any moisture source; air arriving is already dry | Mid-latitude interior deserts such as the Gobi and the Central Asian basins |
| ANALYSIS: Rain shadow | Moist air is forced to rise over a barrier, precipitates on the windward side and descends dry on the lee | Leeward basins behind major ranges; the Patagonian desert behind the Andes |
| ANALYSIS: Cold offshore currents | Air over cold water is chilled from below, becomes stable, and yields fog rather than rain on reaching land | The extreme coastal deserts - Atacama and Namib, both remarkably dry yet foggy |
| ANALYSIS: Polar aridity | Extremely cold air holds very little moisture; subsidence over the ice sheets | The polar cold deserts, treated in `25_■Arctic-■or-■Polar-■Climate.md` |

> MEMORY: **Trap:** "desert" is defined by **moisture deficit**, not by heat or by sand. The largest deserts by area are cold ones, most desert surfaces are **rock and gravel rather than sand**, and the Atacama and Namib are cool and foggy while being among the driest places on Earth.

> MEMORY: **Trap:** more than one cause usually operates at once. The Thar, for example, sits within the subtropical-high belt **and** lies in a position where the summer monsoon current arrives with weak lifting; a single-cause explanation for any large desert is usually wrong.

### 8.2 Desert surfaces: the standard three

| Surface | Character | Why it forms |
|---|---|---|
| ANALYSIS: Erg / sand sea | Extensive dune fields | Where sand supply, wind energy and a depositional basin coincide |
| ANALYSIS: Reg / serir | Stony, gravel-strewn plain, often with a desert pavement | Deflation removes fines and leaves a lag of coarse clasts, which then armours the surface |
| ANALYSIS: Hamada | Bare rock plateau surface | Sweeping removal of loose material, leaving bedrock exposed |

- ANALYSIS: **Sediment transport by wind operates in three modes: suspension** carries the finest dust

far beyond the desert - the source of loess belts elsewhere; **saltation** bounces sand grains and does most of the abrasion; **surface creep** rolls the coarsest grains. This is why wind abrasion is concentrated **near the ground**, producing undercut, pedestal-like rock forms.

### 8.3 Running water in deserts: the process most answers forget

ANALYSIS: Rainfall in deserts is rare, but when it comes it is often intense, falls on sparse vegetation and crusted or bare ground, and generates very high runoff. Over geological time, **episodic water does more to shape most desert landscapes than wind does**.

| Landform | Process |
| --- | --- |
| ANALYSIS: Wadi / arroyo | Steep-sided, normally dry channel cut and periodically re-scoured by flash floods |
| ANALYSIS: Alluvial fan | Cone of sediment where a confined channel loses gradient and confinement at a mountain front |
| ANALYSIS: Bajada | Coalesced belt of alluvial fans along the mountain foot |
| ANALYSIS: Pediment | Gently sloping rock-cut surface at the mountain base, thinly veneered with debris |
| ANALYSIS: Playa / salt pan | Ephemeral lake floor in an interior basin; evaporation concentrates salts |
| ANALYSIS: Inselberg | Isolated residual hill rising abruptly from a pediment plain |

```
MOUNTAIN FRONT
   |  wadi carries flash-flood load out of the confined channel
   v
ALLUVIAL FAN -> merging into a BAJADA
   |  gradient falls further
   v
PEDIMENT (rock-cut) -> BASIN FLOOR
   |  no outlet; water evaporates
   v
PLAYA / SALT PAN, with INSELBERGS standing above the plain
```

> MEMORY: **Trap:** flash floods are the leading cause of death in many desert regions, and wadis are the most dangerous places to camp or build. "Arid" does not mean "no flood risk"; it means the flood is rare, sudden and channel-focused.

### 8.4 Desertification is not the same as desert

- ANALYSIS: A **desert** is a climatic zone. **Desertification** is *land degradation in arid, semi-arid and

dry sub-humid areas*, driven by climatic variation and human activity. A desert cannot "desertify"; the land at risk is the productive dryland margin around it.

- ANALYSIS: Human drivers operate through the same physical mechanisms taught above: removal of vegetation

cover exposes soil to deflation and raindrop impact; overgrazing prevents recovery; cultivation of marginal land breaks surface crusts; and irrigation without drainage salinises the soil (see `04_Weathering-MassMovement-Groundwater.md`).

- ANALYSIS: **The reversibility point:** dryland degradation is often recoverable through vegetation

restoration, shelterbelts, controlled grazing and water harvesting - which is why "advancing desert" imagery is misleading. Institutional and convention-level treatment (UNCCD, land degradation neutrality) belongs to `Environment-and-Ecology`; keep the geomorphic mechanism here.

> ANALYSIS: **Factual caution:** do **not** quote a percentage of India's or the world's land as degraded, a rate of desert advance, or a dune-migration rate from memory.

## 9. Answer architecture (10/15/20-mark support)

### 9.1 Directive decoding for this topic

| If the question says | It is really asking for | Do not |
|---|---|---|
| "Account for the distribution of the world's hot deserts" | The five causes, each matched to a named desert, and the multi-cause qualification | Say "deserts are hot and dry" |
| "Explain aeolian landforms" | Deflation, abrasion and deposition, with the three transport modes and the near-ground abrasion consequence | List dune names only |
| "Distinguish desert from desertification" | Climatic zone versus land-degradation process, plus the reversibility argument | Treat them as synonyms |
| "Discuss the role of water in arid landform development" | The wadi-fan-bajada-pediment-playa sequence and the flash-flood regime | Claim wind does everything |

### 9.2 Reusable 10-mark spine - "Why are deserts located where they are?"

1. **Thesis:** desert location is not accidental; it is the surface expression of atmospheric subsidence, distance from moisture, orographic blocking and ocean-surface temperature - usually in combination.
2. **Cause-by-cause treatment,** each anchored to a named desert (subtropical high -> Sahara; continentality -> Gobi; rain shadow -> Patagonia; cold current -> Atacama and Namib; polar subsidence -> the ice deserts).
3. **Consequence for the landscape:** aridity -> sparse vegetation -> exposed surface -> effective wind action **and** high-runoff flash floods, giving both aeolian and fluvial suites.
4. **Qualification:** the causes overlap; sand is a minority surface; cold deserts are the largest.
5. **Conclusion:** deserts are best read as a **global circulation output**, which is why their margins - not their cores - are where climate variability and human pressure interact most sharply.

### 9.3 Evidence units available in this file

> **Claim:** aridity is created by circulation, not merely by low rainfall totals. **Evidence:** the Atacama and Namib are cool and frequently foggy yet extraordinarily dry, because a cold offshore current stabilises the air and prevents convection. **Significance:** it shows a moisture-bearing ocean can coexist with extreme drought, defeating the intuitive "coast means rain" assumption. **Limitation:** fog does supply some moisture, supporting distinctive fog-dependent ecosystems and even fog-harvesting, so these coasts are not biologically empty.

> **Claim:** the desert margin, not the desert core, is the policy-relevant zone. **Evidence:** degradation processes - deflation of bared soil, overgrazing, marginal cultivation, irrigation salinisation - all operate on the semi-arid fringe where population and cultivation are viable. **Significance:** it redirects an answer from imagery about "spreading sand" to the actual land-management question. **Limitation:** rainfall variability at that margin is itself high, so attributing degradation entirely to human action overstates the case in drought years.

### Semantic-completeness ownership and PYQ control

- **Owned core:** climatic aridity and desert distribution; mechanical and salt weathering; aeolian creep, saltation, suspension, deflation and abrasion; yardangs, ventifacts, pediments, bajadas, wadis, playas and dune types; the Thar's physical mosaic, desertification drivers, canal trade-offs, land-degradation neutrality and process-matched restoration.
- **Process control:** moisture deficit and sparse cover -> weathering and sediment availability -> threshold wind or episodic runoff -> erosion, transport and deposition. Desertification is diagnosed when climatic variability and/or human pressure reduce dryland productivity and recovery, not when a dune simply exists or shifts.
- **Scale/map control:** grain, dune, slope, closed basin, Thar subregion,

dryland class and national atlas pixel are separate scales. Western dune fields, eastern Aravalli transition, Luni-playa system and canal-command tracts cannot be represented by one uniform desert condition.

- **Date/data control:** the SAC atlas reports 2018-19 mapped condition, not a

2026 measurement. Its 97.85 million hectares and 29.77 percent of India's geographical area are baseline-period national values; they do not provide a current Thar-only percentage or prove programme impact.

- **Terminology control:** desert differs from aridity, drought, land

degradation and UNCCD desertification; barchan from transverse, longitudinal, parabolic and star dunes; pediment from bajada; playa from the wider desert; restoration commitment from measured restoration outcome.

- **Causal control:** wind is not the sole geomorphic agent and people are not

inherently a degradation driver. Cover removal, grazing intensity, tillage, groundwater stress, mining, infrastructure, irrigation, drainage and rainfall variability interact differently by site and timescale.

- **Boundary:** Topic 04 owns weathering and groundwater foundations; Topic 18

owns hot- and mid-latitude desert climate; Environment and Ecology owns the direct UNCCD/desertification policy PYQ and Great Indian Bustard conservation. Topic 07 owns arid landforms and the Thar degradation application.

- **Verified PYQ ownership, 2018-2026:** no direct Geography Topic 07 route is

present in the checked central ledgers. The 2020 Mains desertification demand remains with Environment and Ecology and is used only as a bounded cross-owner application; no direct question or objective key is invented.

### GEOGRAPHY DEEP-REVIEW CORE CONTROL

- **Must remember:** Arid geomorphology links moisture deficit, sparse cover, mechanical weathering, episodic runoff, deflation, abrasion and dune migration; pediment-bajada-playa and Thar land-use pressures frame desertification as land degradation.
- **Close distinction:** Desert is a climatic moisture condition, desertification is degradation in drylands, and drought is a temporary anomaly; barchan, transverse, longitudinal and parabolic dunes encode different wind/sediment/vegetation controls.
- **Evidence / causal limit:** Wind is not the sole desert agent and every dune is not freely migrating; grazing, irrigation salinity, groundwater, shelterbelts and rainfall variability have locally different effects, so causal conclusions must be spatially qualified.

## BASIC MCQS / REMEDIATION

### Q1. Which statement correctly explains Concept boundary?

A. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. B. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. C. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. D. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff.

**Answer: A. Explanation: A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. The other options describe different processes, locations, scales or governance categories.**

### Q2. Which option is the safest spatial interpretation of Concept boundary?

A. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. B. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. C. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. D. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering.

**Answer: B. Explanation: A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. The other options describe different processes, locations, scales or governance categories.**

### Q3. Which statement preserves the process boundary for Concept boundary?

A. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. B. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. C. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. D. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces.

**Answer: C. Explanation: A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. The other options describe different processes, locations, scales or governance categories.**

### Q4. Which option avoids the main UPSC trap concerning Concept boundary?

A. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. B. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. C. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. D. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation.

**Answer: D. Explanation: A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. The other options describe different processes, locations, scales or governance categories.**

### Q5. Which statement correctly explains Aridity controls?

A. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. B. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. C. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. D. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces.

**Answer: A. Explanation: Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. The other options describe different processes, locations, scales or governance categories.**

### Q6. Which option is the safest spatial interpretation of Aridity controls?

A. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. B. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. C. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. D. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering.

**Answer: B. Explanation: Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. The other options describe different processes, locations, scales or governance categories.**

### Q7. Which statement preserves the process boundary for Aridity controls?

A. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. B. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. C. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. D. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering.

**Answer: C. Explanation: Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. The other options describe different processes, locations, scales or governance categories.**

### Q8. Which option avoids the main UPSC trap concerning Aridity controls?

A. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. B. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. C. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. D. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient.

**Answer: D. Explanation: Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. The other options describe different processes, locations, scales or governance categories.**

### Q9. Which statement correctly explains Weathering in arid lands?

A. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. B. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. C. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. D. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces.

**Answer: A. Explanation: Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. The other options describe different processes, locations, scales or governance categories.**

### Q10. Which option is the safest spatial interpretation of Weathering in arid lands?

A. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. B. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. C. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. D. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes.

**Answer: B. Explanation: Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. The other options describe different processes, locations, scales or governance categories.**

### Q11. Which statement preserves the process boundary for Weathering in arid lands?

A. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. B. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. C. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. D. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes.

**Answer: C. Explanation: Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. The other options describe different processes, locations, scales or governance categories.**

### Q12. Which option avoids the main UPSC trap concerning Weathering in arid lands?

A. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. B. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. C. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. D. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff.

**Answer: D. Explanation: Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. The other options describe different processes, locations, scales or governance categories.**

### Q13. Which statement correctly explains Aeolian transport?

A. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. B. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. C. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. D. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering.

**Answer: A. Explanation: Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. The other options describe different processes, locations, scales or governance categories.**

### Q14. Which option is the safest spatial interpretation of Aeolian transport?

A. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. B. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. C. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. D. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes.

**Answer: B. Explanation: Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. The other options describe different processes, locations, scales or governance categories.**

### Q15. Which statement preserves the process boundary for Aeolian transport?

A. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. B. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. C. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. D. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods.

**Answer: C. Explanation: Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. The other options describe different processes, locations, scales or governance categories.**

### Q16. Which option avoids the main UPSC trap concerning Aeolian transport?

A. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. B. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. C. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. D. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces.

**Answer: D. Explanation: Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. The other options describe different processes, locations, scales or governance categories.**

### Q17. Which statement correctly explains Erosional landforms?

A. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. B. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. C. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. D. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology.

**Answer: A. Explanation: Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. The other options describe different processes, locations, scales or governance categories.**

### Q18. Which option is the safest spatial interpretation of Erosional landforms?

A. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. B. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. C. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. D. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology.

**Answer: B. Explanation: Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. The other options describe different processes, locations, scales or governance categories.**

### Q19. Which statement preserves the process boundary for Erosional landforms?

A. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. B. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. C. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. D. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe.

**Answer: C. Explanation: Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. The other options describe different processes, locations, scales or governance categories.**

### Q20. Which option avoids the main UPSC trap concerning Erosional landforms?

A. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. B. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. C. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. D. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering.

**Answer: D. Explanation: Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. The other options describe different processes, locations, scales or governance categories.**

### Q21. Which statement correctly explains Dune anatomy?

A. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. B. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. C. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. D. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods.

**Answer: A. Explanation: A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. The other options describe different processes, locations, scales or governance categories.**

### Q22. Which option is the safest spatial interpretation of Dune anatomy?

A. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. B. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. C. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. D. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg.

**Answer: B. Explanation: A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. The other options describe different processes, locations, scales or governance categories.**

### Q23. Which statement preserves the process boundary for Dune anatomy?

A. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. B. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. C. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. D. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods.

**Answer: C. Explanation: A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. The other options describe different processes, locations, scales or governance categories.**

### Q24. Which option avoids the main UPSC trap concerning Dune anatomy?

A. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. B. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. C. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. D. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes.

**Answer: D. Explanation: A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. The other options describe different processes, locations, scales or governance categories.**

### Q25. Which statement correctly explains Water-built desert forms?

A. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. B. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. C. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. D. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg.

**Answer: A. Explanation: Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. The other options describe different processes, locations, scales or governance categories.**

### Q26. Which option is the safest spatial interpretation of Water-built desert forms?

A. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. B. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. C. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. D. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation.

**Answer: B. Explanation: Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. The other options describe different processes, locations, scales or governance categories.**

### Q27. Which statement preserves the process boundary for Water-built desert forms?

A. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. B. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. C. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. D. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe.

**Answer: C. Explanation: Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. The other options describe different processes, locations, scales or governance categories.**

### Q28. Which option avoids the main UPSC trap concerning Water-built desert forms?

A. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. B. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. C. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. D. Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology.

**Answer: D. Explanation: Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. The other options describe different processes, locations, scales or governance categories.**

### Q29. Which statement correctly explains Thar spatial mosaic?

A. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. B. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. C. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. D. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation.

**Answer: A. Explanation: The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. The other options describe different processes, locations, scales or governance categories.**

## Q30. Which option is the safest spatial interpretation of Thar spatial mosaic?

A. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. B. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. C. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. D. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds.

**Answer: B. Explanation: The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. The other options describe different processes, locations, scales or governance categories.**

## Q31. Which statement preserves the process boundary for Thar spatial mosaic?

A. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. B. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. C. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. D. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation.

**Answer: C. Explanation: The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. The other options describe different processes, locations, scales or governance categories.**

## Q32. Which option avoids the main UPSC trap concerning Thar spatial mosaic?

A. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. B. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. C. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. D. The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg.

**Answer: D. Explanation: The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. The other options describe different processes, locations, scales or governance categories.**

## Q33. Which statement correctly explains Thar climate and ecology?

A. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. B. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. C. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. D. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds.

**Answer: A. Explanation: Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. The other options describe different processes, locations, scales or governance categories.**

### Q34. Which option is the safest spatial interpretation of Thar climate and ecology?

A. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. B. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. C. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. D. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds.

**Answer: B. Explanation: Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. The other options describe different processes, locations, scales or governance categories.**

### Q35. Which statement preserves the process boundary for Thar climate and ecology?

A. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. B. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. C. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. D. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak.

**Answer: C. Explanation: Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. The other options describe different processes, locations, scales or governance categories.**

### Q36. Which option avoids the main UPSC trap concerning Thar climate and ecology?

A. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. B. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. C. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. D. Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods.

**Answer: D. Explanation: Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. The other options describe different processes, locations, scales or governance categories.**

### Q37. Which statement correctly explains Origin debate?

A. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. B. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. C. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. D. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement.

**Answer: A. Explanation: The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. The other options describe different processes, locations, scales or governance categories.**

### Q38. Which option is the safest spatial interpretation of Origin debate?

A. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. B. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. C. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. D. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds.

**Answer: B. Explanation: The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. The other options describe different processes, locations, scales or governance categories.**

### Q39. Which statement preserves the process boundary for Origin debate?

A. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. B. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. C. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. D. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement.

**Answer: C. Explanation: The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. The other options describe different processes, locations, scales or governance categories.**

### Q40. Which option avoids the main UPSC trap concerning Origin debate?

A. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. B. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. C. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. D. The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe.

**Answer: D. Explanation: The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. The other options describe different processes, locations, scales or governance categories.**

### Q41. Which statement correctly explains Natural susceptibility?

A. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. B. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. C. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. D. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak.

**Answer: A. Explanation: Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. The other options describe different processes, locations, scales or governance categories.**

### Q42. Which option is the safest spatial interpretation of Natural susceptibility?

A. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. B. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. C. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. D. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation.

**Answer: B. Explanation: Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. The other options describe different processes, locations, scales or governance categories.**

### Q43. Which statement preserves the process boundary for Natural susceptibility?

A. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. B. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. C. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. D. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation.

**Answer: C. Explanation: Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. The other options describe different processes, locations, scales or governance categories.**

### Q44. Which option avoids the main UPSC trap concerning Natural susceptibility?

A. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. B. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. C. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. D. Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation.

**Answer: D. Explanation: Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. The other options describe different processes, locations, scales or governance categories.**

### Q45. Which statement correctly explains Human accelerators?

A. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. B. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. C. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. D. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak.

**Answer: A. Explanation: Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. The other options describe different processes, locations, scales or governance categories.**

### Q46. Which option is the safest spatial interpretation of Human accelerators?

A. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. B. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. C. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. D. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation.

**Answer: B. Explanation: Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. The other options describe different processes, locations, scales or governance categories.**

### Q47. Which statement preserves the process boundary for Human accelerators?

A. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. B. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. C. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. D. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital.

**Answer: C. Explanation: Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. The other options describe different processes, locations, scales or governance categories.**

### Q48. Which option avoids the main UPSC trap concerning Human accelerators?

A. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. B. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. C. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. D. Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds.

**Answer: D. Explanation: Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. The other options describe different processes, locations, scales or governance categories.**

### Q49. Which statement correctly explains Aravalli function?

A. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. B. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. C. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. D. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation.

**Answer: A. Explanation: The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. The other options describe different processes, locations, scales or governance categories.**

### Q50. Which option is the safest spatial interpretation of Aravalli function?

A. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. B. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. C. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. D. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital.

**Answer: B. Explanation: The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. The other options describe different processes, locations, scales or governance categories.**

### Q51. Which statement preserves the process boundary for Aravalli function?

A. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. B. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. C. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. D. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process.

**Answer: C. Explanation: The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. The other options describe different processes, locations, scales or governance categories.**

### Q52. Which option avoids the main UPSC trap concerning Aravalli function?

A. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. B. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. C. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. D. The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement.

**Answer: D. Explanation: The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. The other options describe different processes, locations, scales or governance categories.**

### Q53. Which statement correctly explains Indira Gandhi Canal balance?

A. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. B. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. C. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. D. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital.

**Answer: A. Explanation: Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. The other options describe different processes, locations, scales or governance categories.**

### Q54. Which option is the safest spatial interpretation of Indira Gandhi Canal balance?

A. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. B. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. C. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. D. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process.

**Answer: B. Explanation: Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. The other options describe different processes, locations, scales or governance categories.**

### Q55. Which statement preserves the process boundary for Indira Gandhi Canal balance?

A. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. B. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. C. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. D. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact.

**Answer: C. Explanation: Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. The other options describe different processes, locations, scales or governance categories.**

### Q56. Which option avoids the main UPSC trap concerning Indira Gandhi Canal balance?

A. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. B. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. C. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. D. Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak.

**Answer: D. Explanation: Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. The other options describe different processes, locations, scales or governance categories.**

### Q57. Which statement correctly explains Desertification beyond desert?

A. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. B. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. C. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. D. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact.

**Answer: A. Explanation: The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. The other options describe different processes, locations, scales or governance categories.**

### Q58. Which option is the safest spatial interpretation of Desertification beyond desert?

A. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. B. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. C. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. D. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only.

**Answer: B. Explanation: The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. The other options describe different processes, locations, scales or governance categories.**

### Q59. Which statement preserves the process boundary for Desertification beyond desert?

A. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. B. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. C. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. D. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome.

**Answer: C. Explanation: The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. The other options describe different processes, locations, scales or governance categories.**

### Q60. Which option avoids the main UPSC trap concerning Desertification beyond desert?

A. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. B. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. C. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. D. The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation.

**Answer: D. Explanation: The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. The other options describe different processes, locations, scales or governance categories.**

### Q61. Which statement correctly explains LDN logic?

A. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. B. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. C. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. D. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process.

**Answer: A. Explanation: Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. The other options describe different processes, locations, scales or governance categories.**

### Q62. Which option is the safest spatial interpretation of LDN logic?

A. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. B. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. C. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. D. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only.

**Answer: B. Explanation: Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. The other options describe different processes, locations, scales or governance categories.**

### Q63. Which statement preserves the process boundary for LDN logic?

A. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. B. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. C. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. D. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome.

**Answer: C. Explanation: Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. The other options describe different processes, locations, scales or governance categories.**

### Q64. Which option avoids the main UPSC trap concerning LDN logic?

A. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. B. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. C. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. D. Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital.

**Answer: D. Explanation: Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. The other options describe different processes, locations, scales or governance categories.**

### Q65. Which statement correctly explains Process-matched restoration?

A. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. B. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. C. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. D. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only.

**Answer: A. Explanation: Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. The other options describe different processes, locations, scales or governance categories.**

### Q66. Which option is the safest spatial interpretation of Process-matched restoration?

A. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. B. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. C. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. D. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome.

**Answer: B. Explanation: Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. The other options describe different processes, locations, scales or governance categories.**

### Q67. Which statement preserves the process boundary for Process-matched restoration?

A. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. B. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. C. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. D. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome.

**Answer: C. Explanation: Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. The other options describe different processes, locations, scales or governance categories.**

### Q68. Which option avoids the main UPSC trap concerning Process-matched restoration?

A. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. B. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. C. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. D. Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process.

**Answer: D. Explanation: Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. The other options describe different processes, locations, scales or governance categories.**

### Q69. Which statement correctly explains Monitoring discipline?

A. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. B. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. C. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. D. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome.

**Answer: A. Explanation: Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. The other options describe different processes, locations, scales or governance categories.**

### Q70. Which option is the safest spatial interpretation of Monitoring discipline?

A. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. B. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. C. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. D. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation.

**Answer: B. Explanation: Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. The other options describe different processes, locations, scales or governance categories.**

### Q71. Which statement preserves the process boundary for Monitoring discipline?

A. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. B. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. C. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. D. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff.

**Answer: C. Explanation: Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. The other options describe different processes, locations, scales or governance categories.**

### Q72. Which option avoids the main UPSC trap concerning Monitoring discipline?

A. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. B. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. C. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. D. Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact.

**Answer: D. Explanation: Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. The other options describe different processes, locations, scales or governance categories.**

### Q73. Which statement correctly explains Verified PYQ boundary?

A. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. B. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. C. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. D. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient.

**Answer: A. Explanation: The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. The other options describe different processes, locations, scales or governance categories.**

**Demand decoding:** The directive **answer** requires a direct position on "Q73. Which statement correctly explains Verified PYQ boundary?", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis:** The answer must resolve the physical-geography demand in "Q73. Which statement correctly explains Verified PYQ boundary?".

**Analytical body:**

1. **Claim and named evidence:** Q73. Which statement correctly explains Verified PYQ boundary? **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** B. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** C. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** D. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

**Counter-position / limit:** A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

**Qualified conclusion:** The answer must resolve the physical-geography demand in "Q73. Which statement correctly explains Verified PYQ boundary?".

**Executable exam-length answer / compression plan:** For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

**Why this earns marks:** The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

**How to improve this answer:** For "Q73. Which statement correctly explains Verified PYQ boundary?", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

### Q74. Which option is the safest spatial interpretation of Verified PYQ boundary?

A. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. B. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. C. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. D. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff.

**Answer: B. Explanation: The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. The other options describe different processes, locations, scales or governance categories.**

**Demand decoding:** Treat "Q74. Which option is the safest spatial interpretation of Verified PYQ boundary?" as a process, location, scale, terminology and status problem. Verify every statement independently.

**Detailed examiner-grade model answer:**

**Introduction and thesis:** The answer must resolve the physical-geography demand in "Q74. Which option is the safest spatial interpretation of Verified PYQ boundary?".

**Analytical body:**

1. **Claim and named evidence:** Q74. Which option is the safest spatial interpretation of Verified PYQ boundary? **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** B. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** C. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** D. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

**Counter-position / limit:** A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

**Qualified conclusion:** The answer must resolve the physical-geography demand in "Q74. Which option is the safest spatial interpretation of Verified PYQ boundary?".

**Executable exam-length answer / compression plan:** Fix the phenomenon and spatial setting; trace the controlling mechanism; test direction, sequence, threshold and India/world anchor; eliminate the closest term, map or causation distractor.

**Why this earns marks:** It preserves exact process-to-pattern reasoning and prevents a familiar place-name or correlation from substituting for causation.

**How to improve this answer:** For "Q74. Which option is the safest spatial interpretation of Verified PYQ boundary?", state why the nearest distractor fails on process, direction, scale, location, terminology, date or official status.

## Q75. Which statement preserves the process boundary for Verified PYQ boundary?

A. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. B. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. C. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. D. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff.

**Answer: C. Explanation: The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. The other options describe different processes, locations, scales or governance categories.**

**Demand decoding:** The directive **answer** requires a direct position on "Q75. Which statement preserves the process boundary for Verified PYQ boundary?", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis:** The answer must resolve the physical-geography demand in "Q75. Which statement preserves the process boundary for Verified PYQ boundary?".

**Analytical body:**

1. **Claim and named evidence:** Q75. Which statement preserves the process boundary for Verified PYQ boundary? **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** B. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** C. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** D. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

**Counter-position / limit:** A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

**Qualified conclusion:** The answer must resolve the physical-geography demand in "Q75. Which statement preserves the process boundary for Verified PYQ boundary?".

**Executable exam-length answer / compression plan:** For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

**Why this earns marks:** The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

**How to improve this answer:** For "Q75. Which statement preserves the process boundary for Verified PYQ boundary?", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

### Q76. Which option avoids the main UPSC trap concerning Verified PYQ boundary?

A. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. B. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. C. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. D. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only.

**Answer: D. Explanation: The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. The other options describe different processes, locations, scales or governance categories.**

### Q77. Which statement correctly explains Current evidence boundary?

A. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. B. A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. C. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient. D. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff.

**Answer: A. Explanation: SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. The other options describe different processes, locations, scales or governance categories.**

### Q78. Which option is the safest spatial interpretation of Current evidence boundary?

A. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. B. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. C. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. D. Subtropical subsidence, continentality, rain shadow, cold currents and polar cold can create deserts; heat alone is neither necessary nor sufficient.

**Answer: B. Explanation: SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. The other options describe different processes, locations, scales or governance categories.**

### Q79. Which statement preserves the process boundary for Current evidence boundary?

A. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. B. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. C. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. D. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering.

**Answer: C. Explanation: SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. The other options describe different processes, locations, scales or governance categories.**

### Q80. Which option avoids the main UPSC trap concerning Current evidence boundary?

A. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. B. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. C. A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. D. SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome.

**Answer: D. Explanation: SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. The other options describe different processes, locations, scales or governance categories.**

## PYQS AND ANSWER PRACTICE

**Demand decoding:** Treat "Q76. Which option avoids the main UPSC trap concerning Verified PYQ boundary?" as a process, location, scale, terminology and status problem. Verify every statement independently.

**Detailed examiner-grade model answer:**

**Introduction and thesis:** The answer must resolve the physical-geography demand in "Q76. Which option avoids the main UPSC trap concerning Verified PYQ boundary?".

**Analytical body:**

1. **Claim and named evidence:** Q76. Which option avoids the main UPSC trap concerning Verified PYQ boundary? **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A. Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** B. Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** C. Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

5. **Claim and named evidence:** D. The central routing ledger assigns the 2020 desertification demand to Environment and Ecology, and the checked ledgers contain no direct Geography Topic 07 route; those questions remain cross-owner context only. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

**Counter-position / limit:** A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

**Qualified conclusion:** The answer must resolve the physical-geography demand in "Q76. Which option avoids the main UPSC trap concerning Verified PYQ boundary?".

**Executable exam-length answer / compression plan:** Fix the phenomenon and spatial setting; trace the controlling mechanism; test direction, sequence, threshold and India/world anchor; eliminate the closest term, map or causation distractor.

**Why this earns marks:** It preserves exact process-to-pattern reasoning and prevents a familiar place-name or correlation from substituting for causation.

**How to improve this answer:** For "Q76. Which option avoids the main UPSC trap concerning Verified PYQ boundary?", state why the nearest distractor fails on process, direction, scale, location, terminology, date or official status.

## TRANSPARENT ZERO-DIRECT-PYQ AUDIT

TRANSPARENT ZERO-DIRECT-PYQ AUDIT: the central Mains ledger routes the 2020 desertification demand to Environment and Ecology, and no direct Topic 07 objective route appears in the checked central ledgers. Cross-owned questions remain conceptual context, not claimed Geography PYQs.

## ORIGINAL MAINS 1 - 10 MARKS

**Question:** Differentiate desert, aridity, drought, land degradation and desertification. Answer in about 150 words.

**Model thesis:** The concepts occupy different climatic, temporal and ecological levels; policy fails when they are collapsed.

**Claim -> named evidence -> analysis -> qualification:**

- A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation.
- The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation.

**Qualified conclusion:** The concepts occupy different climatic, temporal and ecological levels; policy fails when they are collapsed.

**Demand decoding:** The directive **answer** requires a direct position on "Differentiate desert, aridity, drought, land degradation and desertification. Answer in about...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis:** The concepts occupy different climatic, temporal and ecological levels; policy fails when they are collapsed.

**Analytical body:**

1. **Claim and named evidence:** A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land

degradation. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

**Counter-position / limit:** A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

**Qualified conclusion:** The concepts occupy different climatic, temporal and ecological levels; policy fails when they are collapsed.

**Executable exam-length answer / compression plan:** For a 10-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise three process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

**Why this earns marks:** The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

**How to improve this answer:** For "Differentiate desert, aridity, drought, land degradation and desertification. Answer in about...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

## ORIGINAL MAINS 2 - 10 MARKS

**Question:** Explain how wind and episodic water jointly create arid landforms. Answer in about 150 words.

**Model thesis:** Aeolian erosion and deposition interact with weathering, flash runoff, fan building, playa flooding and evaporation.

**Claim -> named evidence -> analysis -> qualification:**

- Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff.
- Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces.
- Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering.
- A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes.
- Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology.

**Qualified conclusion:** Aeolian erosion and deposition interact with weathering, flash runoff, fan building, playa flooding and evaporation.

**Demand decoding:** The directive **explain** requires a direct position on "Explain how wind and episodic water jointly create arid landforms. Answer in about 150 words.", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis:** Aeolian erosion and deposition interact with weathering, flash runoff, fan building, playa flooding and evaporation.

**Analytical body:**

1. **Claim and named evidence:** Large thermal ranges, salt crystallisation and episodic wetting weaken exposed rock, while sparse cover leaves debris available to wind and flash runoff. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Wind moves coarse particles mainly by surface creep, sand by saltation and fine dust by suspension; deflation removes loose material and abrasion sandblasts exposed surfaces. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or

observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

3. **Claim and named evidence:** Deflation hollows, desert pavement, ventifacts, yardangs, zeugens, rock pedestals and inselbergs reflect different combinations of wind, structure and weathering. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** A barchan has horns pointing downwind under limited sand and one prevailing wind; transverse, longitudinal, parabolic and star dunes indicate different sand, vegetation and wind regimes. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** Wadis, alluvial fans, bajadas, pediments, playas and salt flats show that episodic runoff and evaporation are integral to desert geomorphology. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

**Counter-position / limit:** A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

**Qualified conclusion:** Aeolian erosion and deposition interact with weathering, flash runoff, fan building, playa flooding and evaporation.

**Executable exam-length answer / compression plan:** For a 10-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise three process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

**Why this earns marks:** The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

**How to improve this answer:** For "Explain how wind and episodic water jointly create arid landforms. Answer in about 150 words.", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

## ORIGINAL MAINS 3 - 15 MARKS

**Question:** Describe the Thar as a dynamic geomorphic and livelihood mosaic rather than a uniform sand sea. Answer in about 250 words.

**Model thesis:** Dunes, pediments, playas, drainage, ecological pulses and human adaptation create regional diversity.

**Claim -> named evidence -> analysis -> qualification:**

- The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg.
- Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods.
- The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe.

**Qualified conclusion:** Dunes, pediments, playas, drainage, ecological pulses and human adaptation create regional diversity.

**Demand decoding:** The directive **describe** requires a direct position on "Describe the Thar as a dynamic geomorphic and livelihood mosaic rather than a uniform sand...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis:** Dunes, pediments, playas, drainage, ecological pulses and human adaptation create regional diversity.

**Analytical body:**

1. **Claim and named evidence:** The Thar includes dune fields, interdunes, rocky residuals, pediments, alluvium, playas, settlements and canal-modified tracts; it is not one continuous erg. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Highly variable rainfall, high evaporative demand, ephemeral drainage and pulse-based productivity support drought-adapted vegetation, pastoral mobility and risk-spreading livelihoods. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** The Thar is polygenetic: tectonic and drainage history, monsoon variability, wind reworking and human land use operated over different timescales; one-cause origin claims are unsafe. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

**Counter-position / limit:** A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

**Qualified conclusion:** Dunes, pediments, playas, drainage, ecological pulses and human adaptation create regional diversity.

**Executable exam-length answer / compression plan:** For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

**Why this earns marks:** The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

**How to improve this answer:** For "Describe the Thar as a dynamic geomorphic and livelihood mosaic rather than a uniform sand...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

## ORIGINAL MAINS 4 - 15 MARKS

**Question:** Assess the role of the Aravalli and the Indira Gandhi Canal in shaping desertification risk. Answer in about 250 words.

**Model thesis:** Both alter spatial processes and livelihoods, but neither is a single-direction barrier or cure.

**Claim -> named evidence -> analysis -> qualification:**

- The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement.
- Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak.

**Qualified conclusion:** Both alter spatial processes and livelihoods, but neither is a single-direction barrier or cure.

**Demand decoding:** The directive **assess** requires a direct position on "Assess the role of the Aravalli and the Indira Gandhi Canal in shaping desertification risk....", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis:** Both alter spatial processes and livelihoods, but neither is a single-direction barrier or cure.

**Analytical body:**

1. **Claim and named evidence:** The Aravalli influences relief, drainage, habitat connectivity and local erosion control, but it is not a complete wall that mechanically blocks all desert movement. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the

directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

2. **Claim and named evidence:** Canal irrigation can support farming, water supply and settlement while seepage, waterlogging, salinity, invasive spread and crop-water mismatch create new risks where drainage is weak. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

**Counter-position / limit:** A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

**Qualified conclusion:** Both alter spatial processes and livelihoods, but neither is a single-direction barrier or cure.

**Executable exam-length answer / compression plan:** For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

**Why this earns marks:** The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

**How to improve this answer:** For "Assess the role of the Aravalli and the Indira Gandhi Canal in shaping desertification risk....", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

## ORIGINAL MAINS 5 - 20 MARKS

**Question:** Evaluate the drivers of Thar desertification and propose a process-matched restoration strategy. Answer in about 300 words.

**Model thesis:** Natural susceptibility becomes degradation when pressures exceed recovery; responses must target cover, soil, water, salinity and institutions.

**Claim -> named evidence -> analysis -> qualification:**

- Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation.
- Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds.
- Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process.
- Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact.

**Qualified conclusion:** Natural susceptibility becomes degradation when pressures exceed recovery; responses must target cover, soil, water, salinity and institutions.

**Demand decoding:** The directive **evaluate** requires a direct position on "Evaluate the drivers of Thar desertification and propose a process-matched restoration...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis:** Natural susceptibility becomes degradation when pressures exceed recovery; responses must target cover, soil, water, salinity and institutions.

**Analytical body:**

1. **Claim and named evidence:** Low and variable rainfall, erodible sediment, strong winds, saline basins and slow biological recovery create susceptibility without proving degradation. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Vegetation removal, poorly managed grazing, repeated tillage, groundwater stress, mining, infrastructure fragmentation and unsuitable irrigation can exceed recovery thresholds. **Analysis:** Trace the

driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

3. **Claim and named evidence:** Dune stabilisation, shelterbelts, native grass and rangeland recovery, watershed treatment, drainage, salinity management and community institutions must match the diagnosed process. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** Land-cover change, vegetation productivity, soil carbon and field indicators need scale, baseline and causal interpretation; mapped degradation is not automatically current condition or programme impact. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

**Counter-position / limit:** A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

**Qualified conclusion:** Natural susceptibility becomes degradation when pressures exceed recovery; responses must target cover, soil, water, salinity and institutions.

**Executable exam-length answer / compression plan:** For a 20-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise six to eight process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

**Why this earns marks:** The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

**How to improve this answer:** For "Evaluate the drivers of Thar desertification and propose a process-matched restoration...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

## ORIGINAL MAINS 6 - 20 MARKS

**Question:** Justify the proposition that desertification does not have simple climatic boundaries while retaining UNCCD precision. Answer in about 300 words.

**Model thesis:** Use broad degradation mechanisms across climates, then explicitly preserve the Convention's dryland definition.

**Claim -> named evidence -> analysis -> qualification:**

- A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation.
- The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation.
- Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital.
- SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome.

**Qualified conclusion:** Use broad degradation mechanisms across climates, then explicitly preserve the Convention's dryland definition.

**Demand decoding:** The directive **answer** requires a direct position on "Justify the proposition that desertification does not have simple climatic boundaries while...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis:** Use broad degradation mechanisms across climates, then explicitly preserve the Convention's dryland definition.

**Analytical body:**

1. **Claim and named evidence:** A desert is a climatic region, aridity a long-term moisture deficit, drought a temporary anomaly, land degradation a broad decline in condition, and desertification dryland degradation. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** The 2020 GS-I wording invites examples beyond visible deserts, but strict UNCCD usage confines desertification to arid, semi-arid and dry sub-humid lands; humid cases are broader land degradation. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Land Degradation Neutrality seeks to balance anticipated losses with measures that avoid, reduce and reverse degradation while protecting land-based natural capital. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** SAC's Desertification Status Mapping Atlas is an official dated mapping baseline and UNCCD's 2026 material keeps restoration current; neither licenses an invented current Thar percentage or COP outcome. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

**Counter-position / limit:** A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

**Qualified conclusion:** Use broad degradation mechanisms across climates, then explicitly preserve the Convention's dryland definition.

**Executable exam-length answer / compression plan:** For a 20-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise six to eight process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

**Why this earns marks:** The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

**How to improve this answer:** For "Justify the proposition that desertification does not have simple climatic boundaries while...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

## OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

> **Subject:** Geography · **Tier:** Advanced (India-applied) · **GS Paper:** GS-I **Grounded in:** D.R. Khullar, *India: A Comprehensive Geography* + Majid Husain, *India geography* + verified CA. FACT: = from source book · ANALYSIS: = inference / standard knowledge · CURRENT: = current affairs. *Companion:* `basic/07_Arid-Desert-Landforms.md`.

### 1. Snapshot & core idea

**Advanced - The Thar (Great Indian Desert): Extent & Aridity (Khullar / Majid).**

FACT: Khullar places the arid/semi-arid Thar chiefly west of the Aravalli in Rajasthan, extending into adjoining Punjab, Haryana, Gujarat and Pakistan. Older regional descriptions link it with the Rann of Kachchh, but the Rann is a **distinct saline mudflat/marsh system**, not simply another field of Thar dunes. The sand derives from mechanical disintegration of local rocks (and blown-in material). West of the Aravalli - Jaisalmer and Barmer - is classic aeolian topography dominated by sand dunes.

- FACT: The Thar lies west of the Aravalli, between the Indus and the Aravalli, spilling into Punjab, Haryana and northern Gujarat.
- FACT: Book estimates of desert area vary with the boundary used; retain them as source-era figures, not

as a single current official area. Rainfall grades from semi-arid margins to an arid western core.

- FACT: The Rann of Kachchh is treated as an extension of the desert; Jaisalmer and Barmer show the strongest aeolian (dune) topography.
- FACT: Economy (Majid): bajra, pulses, fodder; livestock is central; scope for date-palm, water-melon and rain-water harvesting.
- FACT: It is described as the Western Dry Region - an arid tract (Marwar/Mewar margins) with extreme temperature range.

**Ca Anchor - ISRO-SAC Desertification & Land Degradation Atlas.**

FACT: Desertification is not just sand spreading - it is loss of dryland productivity by water/wind erosion and vegetation loss, mapped nationally to guide restoration.

- FACT: ~29.7% of India (~97.85 mha) is degraded (2018-19), up from 28.76% (2003-05) - a rising trend.
- FACT: Rajasthan has the largest desertified area; wind erosion drives 40%+ of its desertification.
- FACT: India targets Land Degradation Neutrality (LDN) by 2030 under the UNCCD; tools include Green India Mission and watershed/afforestation programmes.

## 2. Key classification / data

| Attribute | Value / detail | Source significance |
|---|---|---|
| FACT: Location | West of Aravalli; Rajasthan with adjoining Punjab/Haryana; between Indus and Aravalli | Defines the Great Indian Desert setting |
| FACT: Area | Varies with whether only the arid core or wider semi-arid region is included | Always state delimitation/source |
| FACT: Rainfall | <50 cm annually in Khullar note; <25 cm arid core in source note | Aridity control |
| FACT: Rann relationship | Kachchh, Gujarat | Arid-region linkage, but geomorphically a saline mudflat/marsh distinct from sandy Thar |
| FACT: Aeolian core | Jaisalmer and Barmer | Strong sand-dune topography |
| FACT: Economy | Bajra, pulses, fodder, livestock, date-palm possibilities | Livelihood adaptation |
| FACT: Degradation driver | Wind erosion, unsuitable cultivation, poor moisture conservation and overgrazing | Source-book dryland analysis |
| FACT: Conservation direction | Agroforestry, pasture management, water conservation and reclamation | Source-book measures, updated with landscape-restoration logic |

## 3. Study links

> **Study link:** FACT: Geography -> India -> Arid Region / Western Dry Region (Khullar; Majid: Physiography) **Study link:** FACT: Geography -> Arid India -> Desertification & LDN (applied CA)

## 4. Must-Know Facts (Prelims)

**Advanced - The Thar (Great Indian Desert): Extent & Aridity (Khullar / Majid).**

- FACT: Thar = west of Aravalli, between Indus & Aravalli.
- FACT: Area figures vary by delimitation; western core is arid and margins semi-arid.
- FACT: Rann of Kachchh is linked to the wider arid region but is a distinct saline mudflat/marsh.
- FACT: Jaisalmer & Barmer = strongest aeolian (dune) topography.
- FACT: Core crops: bajra, pulses, fodder; livestock-based economy.

**Ca Anchor - ISRO-SAC Desertification & Land Degradation Atlas.**

- FACT: Atlas by ISRO-SAC; ~29.7% of India degraded (2018-19).
- FACT: Rajasthan = largest desertified area (wind erosion dominant).
- FACT: India target: Land Degradation Neutrality by 2030 (UNCCD).

### 5. UPSC Traps

> MEMORY: Trap: Do not confuse similar-looking landforms, processes, scales or Indian locations; UPSC often tests the exact mechanism and setting.

**Advanced - The Thar (Great Indian Desert): Extent & Aridity (Khullar / Majid).**

- WRONG: The Thar lies east of the Aravalli. -> The Thar is WEST of the Aravalli range.
- WRONG: The Rann of Kachchh is unrelated to the Thar. -> Khullar treats the Rann of Kachchh as an EXTENSION of the Thar desert.
- WRONG: The Thar receives moderate rainfall (~75 cm). -> It receives < 50 cm (Khullar); the arid core is < 25 cm (Majid).

**Ca Anchor - ISRO-SAC Desertification & Land Degradation Atlas.**

- WRONG: Desertification means only the spread of sand dunes. -> It means loss of dryland PRODUCTIVITY (water/wind erosion, vegetation loss) - often reversible.

### 6. CURRENT: Current link

**Advanced - The Thar (Great Indian Desert): Extent & Aridity (Khullar / Majid).** CURRENT: The Thar's current significance lies in the contrast between mobile/stabilised dunes, canal-irrigated tracts, renewable-energy expansion and land degradation. Quote area and rainfall only with the source boundary/year.

**Ca Anchor - ISRO-SAC Desertification & Land Degradation Atlas.** CURRENT: ISRO's Space Applications Centre 'Desertification & Land Degradation Atlas of India' shows ~29.7% of India's land degraded (2018-19), with Rajasthan the largest desertified area - mostly by wind erosion.

### 7. Mains angles

**Advanced - The Thar (Great Indian Desert): Extent & Aridity (Khullar / Majid).**

- ANALYSIS: Characterise the Thar's extent, aridity and aeolian landscape, and link its physical setting to livelihood strategies and desertification risk.

**Ca Anchor - ISRO-SAC Desertification & Land Degradation Atlas.**

- ANALYSIS: Use the Atlas to argue for targeted, region-specific land-restoration to meet India's 2030 LDN goal.

## CONSOLIDATED REGISTER NOTES

### COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM

## ASCII MASTER FLOW - PANEL 1/12: Concept firewall

DESERT -> climatic region
ARIDITY -> long-term moisture deficit
DROUGHT -> temporary anomaly
DESERTIFICATION -> degradation within drylands
MUST REMEMBER: Arid geomorphology links moisture deficit, sparse cover, mechanical weathering, episodic runoff, deflation, abrasion and dune migration; pediment-bajada-playa and Thar land-use pressures frame desertification as land degradation.
MUST REMEMBER: Arid geomorphology links moisture deficit, sparse cover, mechanical weathering, episodic runoff, deflation, abrasion and dune migration; pediment-bajada-playa and Thar land-use pressures frame desertification as land degradation.

## ASCII MASTER FLOW - PANEL 2/12: Desert-forming controls

SUBTROPICAL HIGH -> descending dry air
CONTINENTALITY / RAIN SHADOW -> moisture loss
COLD CURRENT -> stable dry coastal air
POLAR COLD -> low absolute moisture

## ASCII MASTER FLOW - PANEL 3/12: Aeolian transport

CREEP -> coarse grains roll or slide
SALTATION -> sand grains bounce
SUSPENSION -> fine dust travels far
DEPOSITION -> velocity or carrying power falls

## ASCII MASTER FLOW - PANEL 4/12: Dune discriminator

BARCHAN -> limited sand, horns downwind
TRANSVERSE -> abundant sand, one dominant wind
LONGITUDINAL -> aligned with resultant winds
PARABOLIC / STAR -> vegetation anchor / multidirectional winds

## ASCII MASTER FLOW - PANEL 5/12: Water in the desert

STORM RUNOFF -> wadi discharge
MOUNTAIN FRONT -> alluvial fan
FANS COALESCE -> bajada
CLOSED BASIN -> playa and salt concentration

## ASCII MASTER FLOW - PANEL 6/12: Thar spatial anatomy

WEST -> sandy Marusthali core
EAST -> Bagar and Aravalli transition
SOUTHWEST -> Luni and saline-basin route
NORTH -> canal-alluvial transformation

## ASCII MASTER FLOW - PANEL 7/12: Susceptibility-to-degradation

```
VARIABLE RAIN + WIND -> natural stress
COVER LOSS + SOIL DISTURBANCE -> exposure
WATER MISMANAGEMENT -> salinity or depletion
THRESHOLD CROSSED -> recovery capacity declines
```

## ASCII MASTER FLOW - PANEL 8/12: Aravalli boundary

```
RELIEF -> modifies local wind and runoff
DRAINAGE -> divides and routes water
HABITAT -> supports connectivity
LIMIT -> not an impermeable desert wall
```

## ASCII MASTER FLOW - PANEL 9/12: Canal balance

```
BENEFIT -> water supply and production
SEEPAGE -> water-table rise
POOR DRAINAGE -> waterlogging and salinity
MANAGEMENT -> crop-water fit + drainage + monitoring
```

## ASCII MASTER FLOW - PANEL 10/12: LDN response hierarchy

```
AVOID -> protect functioning land
REDUCE -> improve grazing, tillage and water use
REVERSE -> restore soil, grass and drainage
BALANCE -> monitor gains and losses transparently
CLOSE DISTINCTION: Desert is a climatic moisture condition, desertification is degradation in
  drylands, and drought is a temporary anomaly; barchan, transverse, longitudinal and
  parabolic dunes encode different wind/sediment/vegetation controls.
```

```
CLOSE DISTINCTION: Desert is a climatic moisture condition, desertification is degradation in
  drylands, and drought is a temporary anomaly; barchan, transverse, longitudinal and
  parabolic dunes encode different wind/sediment/vegetation controls.
```


## ASCII MASTER FLOW - PANEL 11/12: Restoration matrix

```
WIND EROSION -> cover, shelter and dune management
RUNOFF EROSION -> contour and watershed treatment
SALINITY -> drainage and water balance
RANGELAND DECLINE -> native grass + institutions
EVIDENCE LIMIT: Wind is not the sole desert agent and every dune is not freely migrating;
  grazing, irrigation salinity, groundwater, shelterbelts and rainfall variability have
  locally different effects, so causal conclusions must be spatially qualified.
```

```
EVIDENCE LIMIT: Wind is not the sole desert agent and every dune is not freely migrating;
  grazing, irrigation salinity, groundwater, shelterbelts and rainfall variability have
  locally different effects, so causal conclusions must be spatially qualified.
```

## ASCII MASTER FLOW - PANEL 12/12: Desertification answer spine

```
DEFINE -> retain UNCCD dryland boundary
MAP -> locate process and pressure
DIAGNOSE -> susceptibility plus accelerator
RESPOND -> process-matched, monitored and qualified
```